A Toy Model for Testing Asset Extraction

Ellrin Example and Daidai Sample

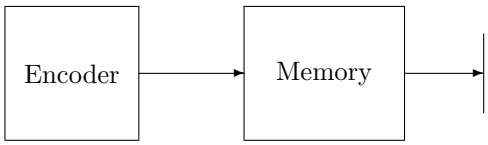


Figure 1: The toy architecture with an encoder and a memory block.

Abstract

We describe a toy encoder and decoder pipeline that exists only to test the extraction of figures, tables and equations from a two-column paper.

1 Introduction

Toy pipelines are useful for testing because every element is known in advance. The encoder reads an image and the decoder writes a short report, as shown in Fig. 1. We summarise the results in Table 1 and define the loss in Eq. (2). This paragraph continues with ordinary prose so that the column has enough body text to bound the regions above and below each float, which is what the detector relies on when it searches for the free space next to a caption.

2 Method

The encoder maps an input x to features h with a single linear layer, and the decoder reads the features together with a small memory bank M that stores report fragments for later use.

$$h = W x + b \tag{1}$$

The memory is updated with a running average of the features seen during training, which keeps the toy model stable.

$$\mathcal{L} = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{2}$$

where y_i is the target and $\hat{y}_i$ is the prediction. Training uses plain gradient descent with a fixed learning rate and no tricks at all, so that the text stays simple for the parser.

3 Results

Table 1 lists the scores of the toy model and a baseline on two invented datasets. The toy model is better on both, which is not surprising since the numbers were chosen by hand for this test document.

Table 1: Scores on the invented datasets.

Method	Set A	Set B
Baseline	0.61	0.58
Toy model	0.72	0.69

More prose follows the table so that its region is bounded below by body text. The discussion is short because the document has no real findings to discuss beyond the behaviour of the extractor itself.

References

[1] E. Example, “A first toy reference,” *Journal of Tests*, 2020.

[2] D. Sample, “A second toy reference,” in *Proc. Imaginary Conf.*, 2021.